

Temperature Sensors and Motors

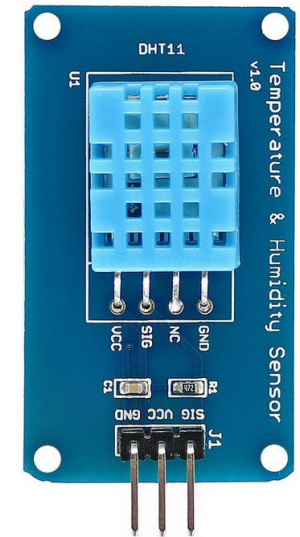
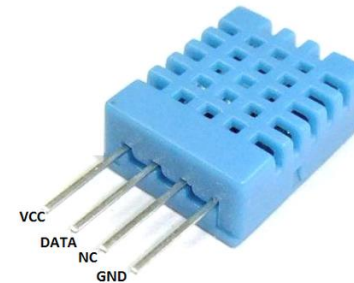
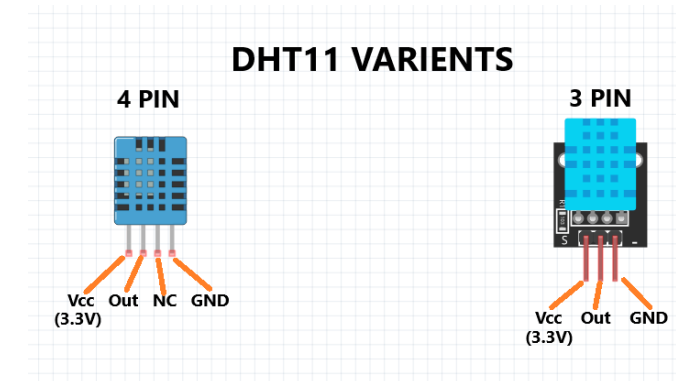
Agenda

- Environmental Sensors
- Motors

Temperature and Humidity

DHT11 : The DHT11 is a basic, ultra low-cost digital temperature and humidity sensor. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air.

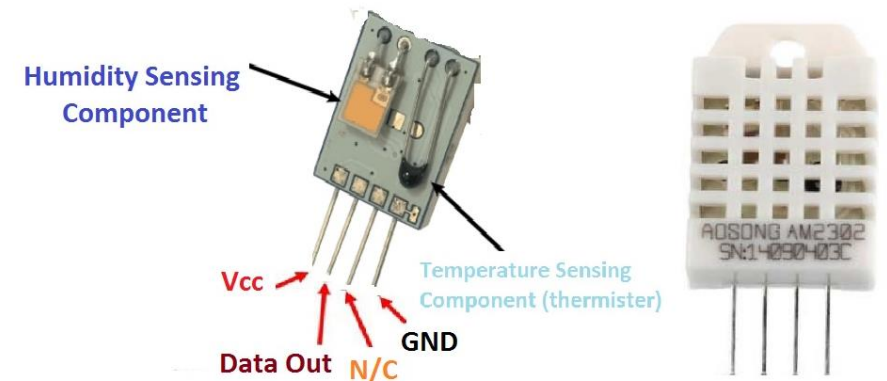
- 0-50 °C temperature readings ± 2 °C accuracy
- 20-80% humidity readings with 5% accuracy



Temperature and Humidity (Cont.)

The DHT22 is a basic digital temperature and humidity sensor. With a normal and not real-time response time.

- -40 to 80°C temperature readings $\pm 0.5^\circ\text{C}$ accuracy
- 0-100% humidity readings with 2-5% accuracy



DS18B20 Digital temperature sensor

This temperature sensor has a digital output, which works well with the Raspberry Pi.

Handy for when you need to measure something far away, or in wet conditions.

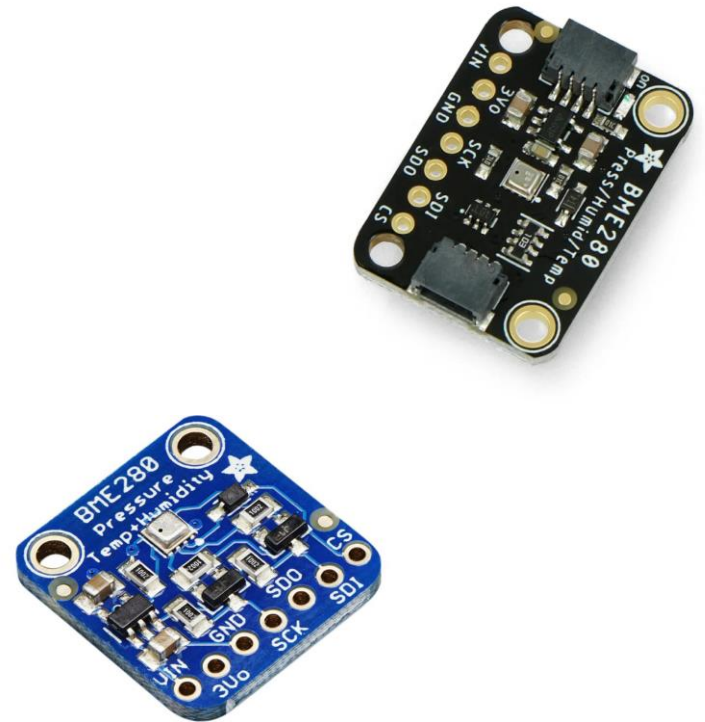
-55 to 125°C temperature readings $\pm 0.5^\circ\text{C}$ accuracy

Response time is less than 750ms

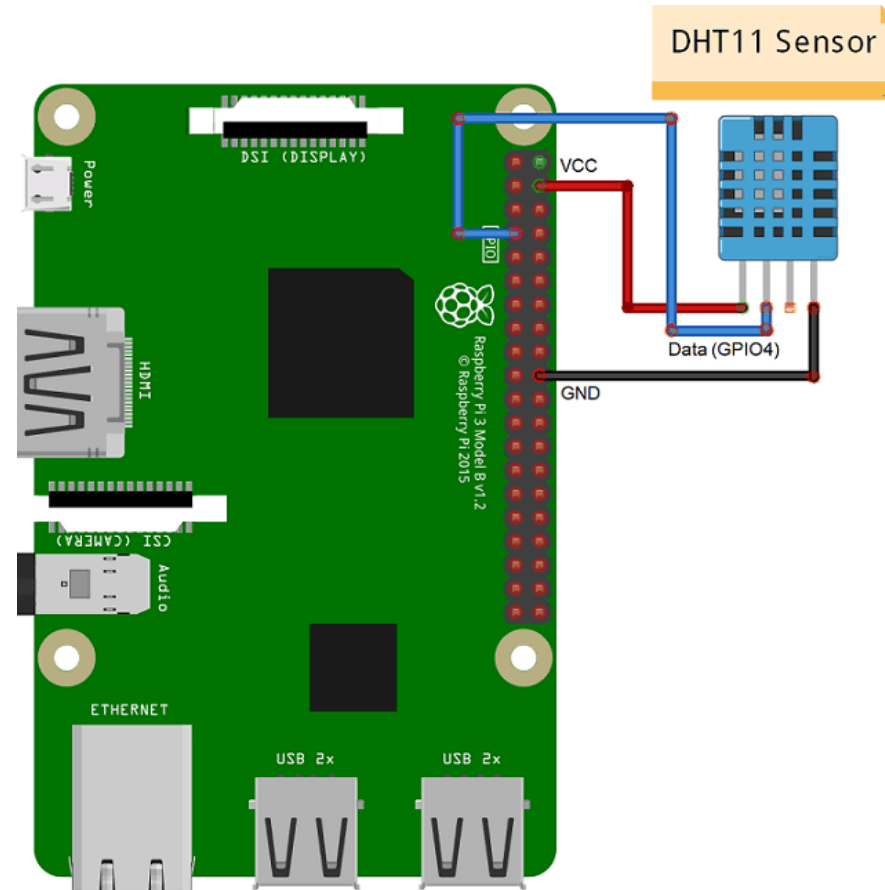


Bosch sensor

- This is an Air Pressure and Temperature and Humidity sensor from Bosch Sensortec.
- This precision sensor from Bosch is the best low-cost sensing solution for measuring humidity with $\pm 3\%$ accuracy, barometric pressure with ± 1 hPa absolute accuracy, and temperature with $\pm 1.0^{\circ}\text{C}$ accuracy.



Temperature sensor circuit



Motors

- When planning a motion control system, the choice of motor is extremely important. The motor must align with the purpose and overall performance goals of the system.
- Three common motors:
 - DC
 - Stepper
 - Servo



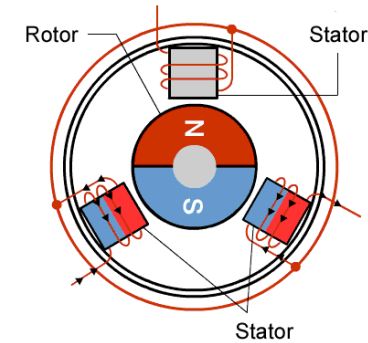
DC Motors

DC (Direct Current) Motors are two wire (power & ground), continuous rotation motors.

Electromagnetic devices that use the interaction of magnetic fields and conductors to convert electrical energy to mechanical energy for rotation.

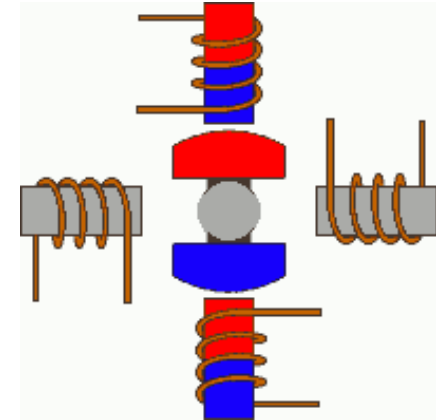
There are many types of DC motors out in the market.

The brushed and brushless motors are the most common DC motors.



Stepper motors

- Stepper motors are motors that move in slow, precise and discrete steps.
- Valued for their precise position control, they find a numerous of applications such as desktop printers, security cameras, and CNC milling machines

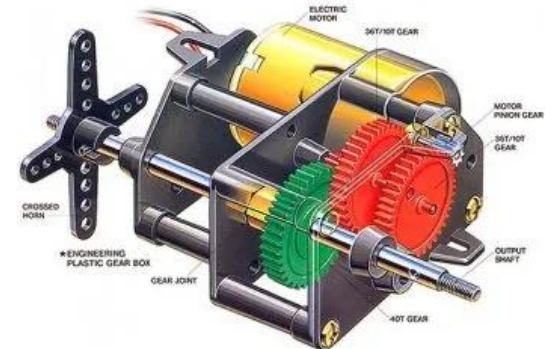


Servo motors

Servo motors are motors capable of providing very precise motion control.

Servo motors are generally an assembly of four things: a DC motor, a gearing set, a control circuit and a position-sensor.

Usually servo motor rotates within a 180 degrees range.

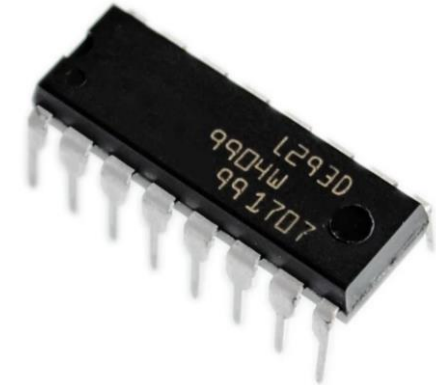


Motor Driver IC

A motor driver IC is an integrated circuit chip which is usually used to control motors in autonomous robots.

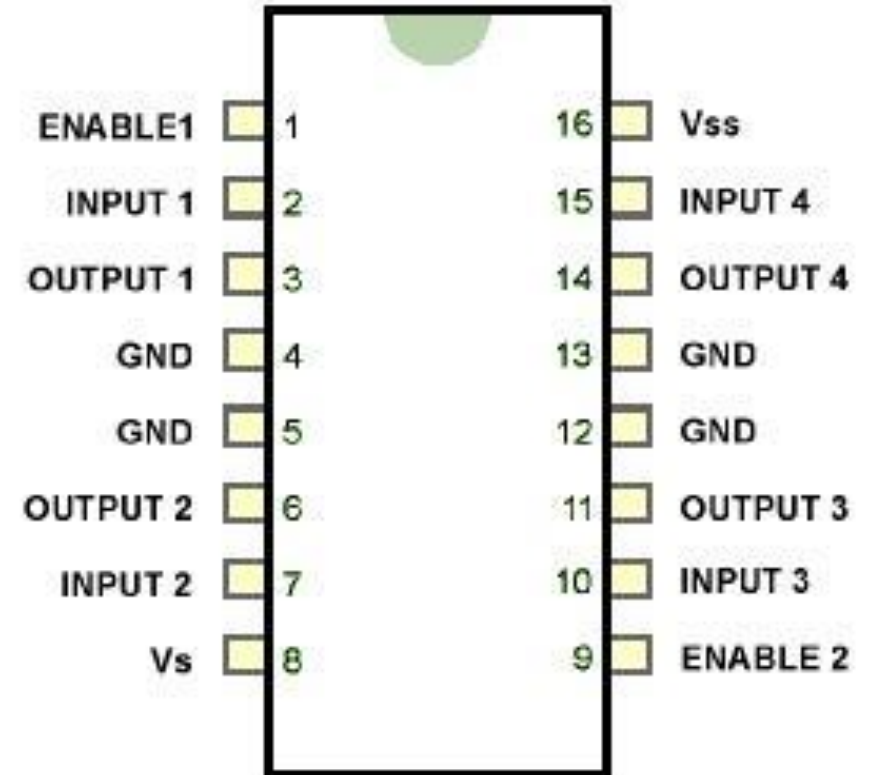
Motor driver ICs act as an interface between microprocessors and the motors.

Current cannot be supplied to the motors from the microprocessor. This is the primary need for the motor driver IC.



L293D motor driver

- The L293D IC receives signals from the microprocessor and transmits the relative signal to the motors.
- It is a 16 pin IC, with eight pins, on each side, dedicated to the controlling of a motor.



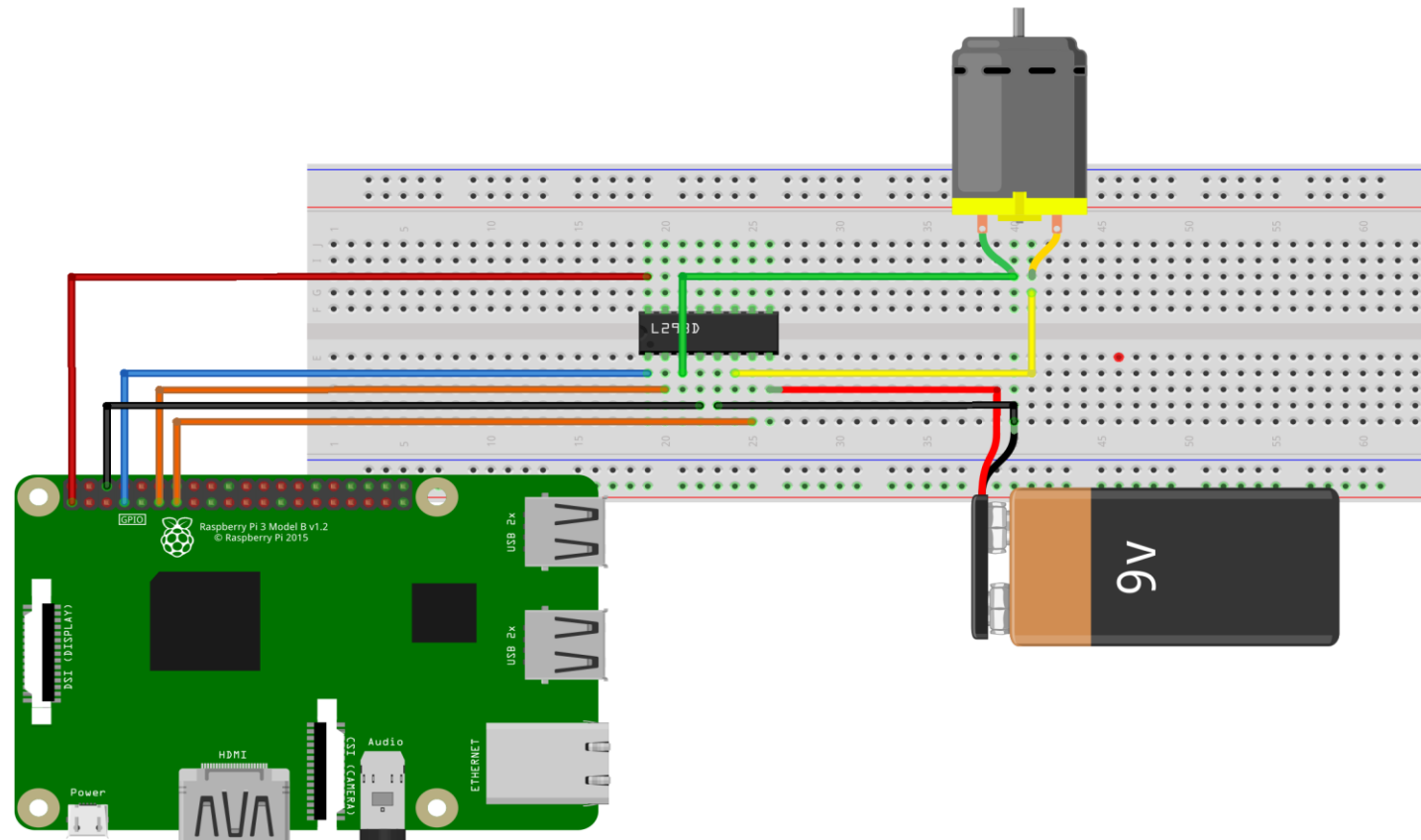
L293D Working Mechanism

Depending upon the values of the Input and Enable the motors will rotate in either clockwise or anticlockwise direction.

Input1	Input2	Enable	Action
x	x	0	No-Action
0	0	1	Stop
0	1	1	Counter-clockwise
1	0	1	Clockwise
1	1	1	Stop

DC motor circuit

Red (Dark)= RPi Vcc
Red (Light)= Motor Vcc
Orange = Input
Blue = Enable
Black = Ground
Green = Out-1
Yellow = Out-2



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DC Motor Code

```
import RPi.GPIO as GPIO  
from time import sleep  
GPIO.setwarnings(False)  
GPIO.setmode(GPIO.BCM)  
enable=12  
pin1=26  
pin2=21  
GPIO.setup(enable, GPIO.OUT)  
GPIO.setup(pin1, GPIO.OUT)  
GPIO.setup(pin2, GPIO.OUT)
```

DC Motor Code (Cont.)

```
GPIO.output(pin1, 1)
```

```
GPIO.output(pin2, 0)
```

```
pwm=GPIO.PWM(enable, 100)
```

```
pwm.start(0)
```

```
pwm.ChangeDutyCycle(50)
```

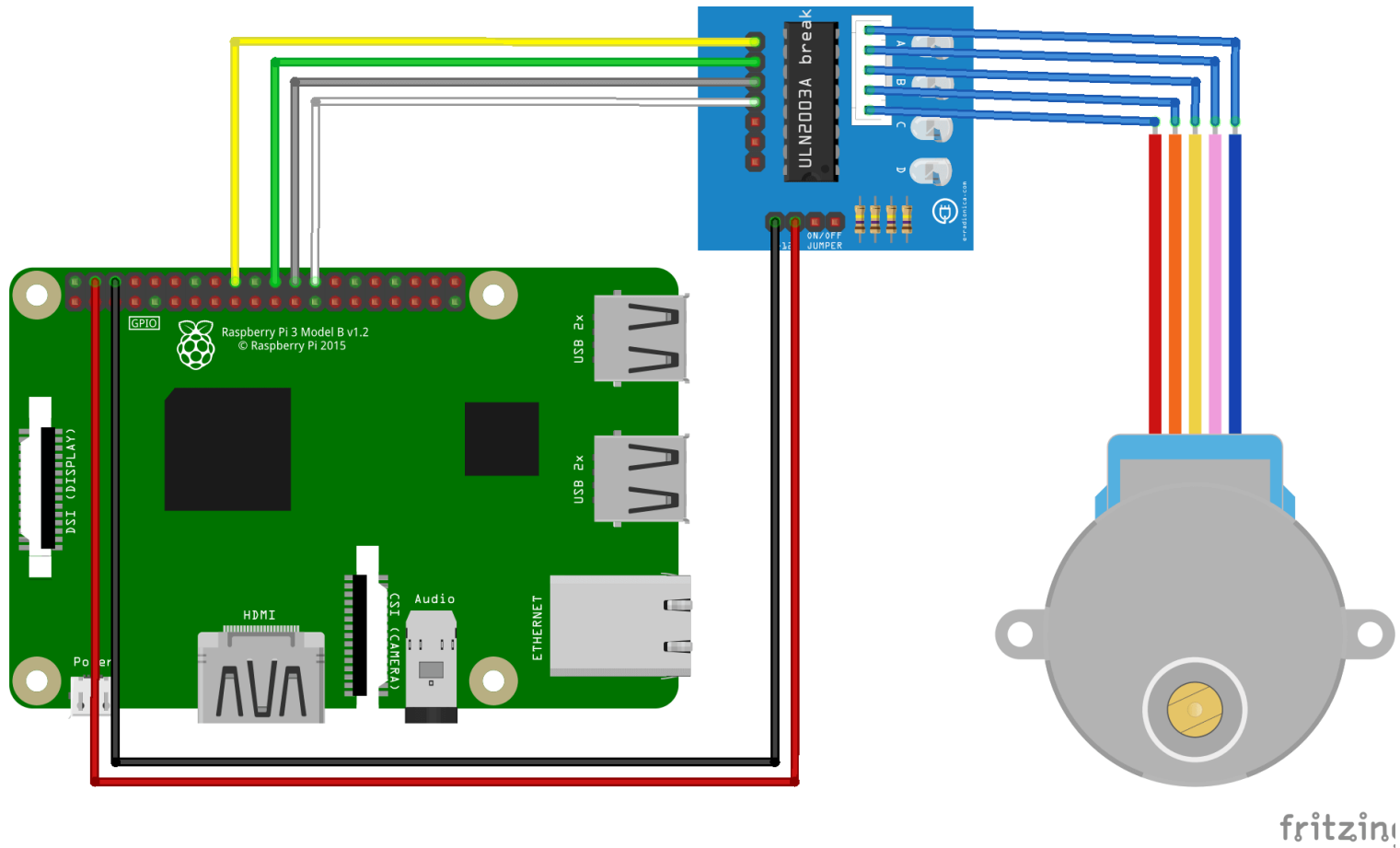
```
GPIO.output(enable, 1)
```

```
sleep(5)
```

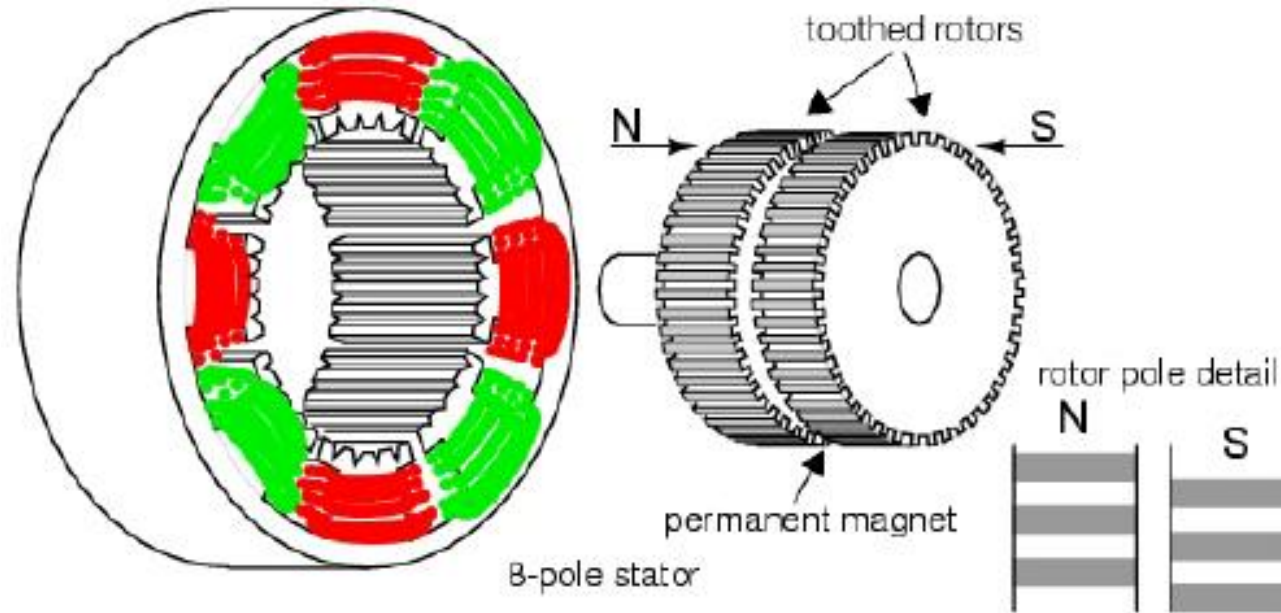
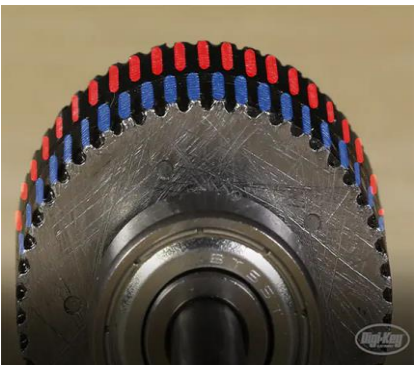
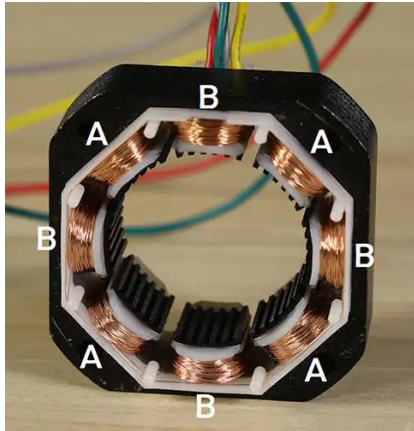
```
GPIO.output(enable, 0)
```

```
GPIO.cleanup()
```

Stepper Motor Circuit



Stepper Motor Structure



Stepper Motor Code

```
import RPi.GPIO as GPIO
```

```
import time
```

```
GPIO.setmode(GPIO.BOARD)
```

```
GPIO.setwarnings(False)
```

```
ControlPin = [7,11,13,15]
```

```
for pin in ControlPin:
```

```
    GPIO.setup(pin, GPIO.OUT)
```

```
    GPIO.output(pin,0)
```

Stepper Motor Code (Cont.)

$$\begin{aligned} seq = & [[1,0,0,0], \\ & [1,1,0,0], \\ & [0,1,0,0], \\ & [0,1,1,0], \\ & [0,0,1,0], \\ & [0,0,1,1], \\ & [0,0,0,1], \\ & [1,0,0,1]] \end{aligned}$$

Stepper Motor Code (Cont.)

```
angle = float(input('Enter angle between 0 & 360 '))  
an= int(angle*1.42)  
for i in range(an):  
    for halfstep in range(8):  
        for pin in range(4):  
            GPIO.output(ControlPin[pin], seq[halfstep][pin])  
            time.sleep(0.001)  
GPIO.cleanup()
```